

E-commerce Funnel & Revenue Optimization

Analysis

1. Problem Statement

The objective of this analysis is to identify inefficiencies in the e-commerce conversion funnel and evaluate how different traffic sources contribute to revenue.

Although the business is generating a steady flow of traffic, there is limited visibility into how users move through the funnel and where drop-offs occur. Without this understanding, marketing efforts may be misaligned, leading to inefficient spending and lost revenue opportunities.

2. Dataset Description

The dataset captures user interactions across different stages of the e-commerce funnel.

Key Columns include:

- user_id – Unique identifier for each user
- event_type – Represents funnel stage (page_view, add_to_cart, checkout, payment, purchase)
- event_date – Timestamp of the user action
- amount – Transaction value for purchases
- traffic_source – Channel through which the user was acquired (organic, paid ads, email, social)

Data Preparation steps:

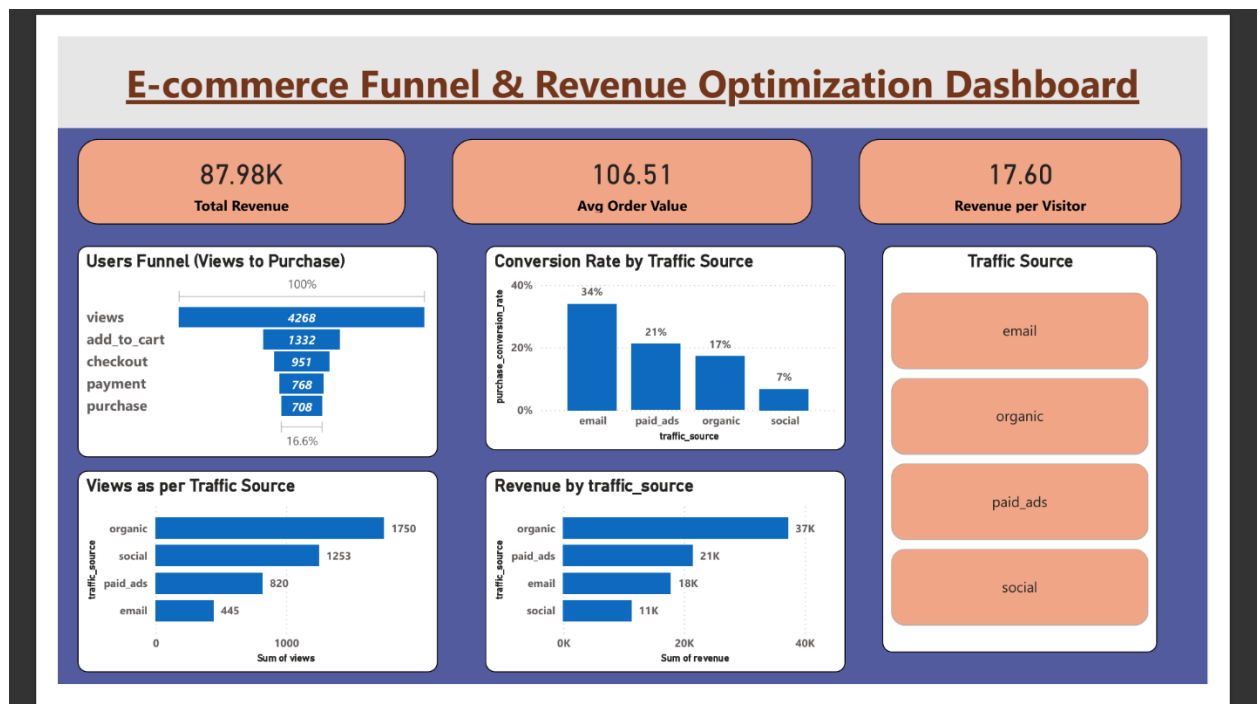
- Cleaned null/blank values in amount
- Converted data types for accurate calculations
- Filtered analysis to recent 30-day period

3. Methodology

The analysis was conducted in the following steps:

- Cleaned and standardized the dataset
- Constructed the user funnel using event-based stages
- Calculated conversion rates at each stage
- Analyzed performance across traffic sources
- Evaluated revenue metrics such as total revenue, AOV, and revenue per visitor
- Built a dashboard to visualize key insights

4. Dashboard Overview



The dashboard provides a consolidated view of user behavior, conversion performance, and revenue metrics.

Key performance indicators show total revenue of 87.98K, an average order value of 106.51, and revenue per visitor of 17.60, giving a clear snapshot of overall business performance.

The funnel highlights a drop from 4,268 users at the page view stage to 708 final purchases, resulting in an overall conversion rate of approximately 16.6%. The largest decline occurs at the initial stage (view to add-to-cart), indicating weak early engagement.

Traffic source analysis shows that email has the highest conversion rate (~34%) despite low volume, while social traffic performs the worst (~7%), suggesting low-quality traffic acquisition.

Organic traffic generates the highest revenue (~37K), making it the most valuable channel in terms of contribution.

5. Key Analysis & Findings

1. Funnel Drop-Off (Critical Insight)

A significant drop (Views: 4268 → Purchases: 708) occurs at the beginning of the funnel, indicating that users are not progressing beyond the initial interaction stage.

Interpretation:

This suggests issues with product presentation, targeting, or user intent at the top of the funnel.

2. Strong Checkout Performance

- Minimal drop from **Payment** → **Purchase**

Interpretation:

Checkout process is efficient and does not require optimization.

3. Traffic Source Performance

- **Organic** → Highest traffic (1750 views) and highest revenue (~37K)
- **Paid Ads** → Moderate performance (~21K revenue)
- **Email** → Low traffic but **highest conversion rate (34%)**
- **Social** → High traffic but **lowest conversion (~7%) and lowest revenue**

- **Interpretation:**

Traffic quality varies significantly across channels.

4. Revenue Insights

- Total Revenue: **87.98K**
- Average Order Value: **106.51**
- Revenue per Visitor: **17.60**

Revenue is concentrated in a few high-performing channels, while a large portion of traffic contributes minimally.

Interpretation:

This indicates inefficiency in how traffic is being acquired and utilized.

6. Business Insights

- The primary issue lies in early-stage engagement rather than checkout
- Email marketing attracts high-intent users
- Social media drives traffic but fails to convert
- High traffic does not necessarily translate into high revenue

7. Recommendations

1. Improve Top-of-Funnel Engagement

- Optimize product pages, CTAs, and UX
- Test better product visuals and descriptions

2. Reduce Social Media Spend

- Shift from traffic campaigns → retargeting campaigns
- Focus on quality over quantity

3. Scale Email Marketing

- Implement email capture strategies (popups, offers)

- Leverage remarketing campaigns

4. Monitor Marketing Efficiency

- Align Customer Acquisition Cost (CAC) with AOV (~106)
- Avoid overspending on low-converting channels

8. Limitations

- No segmentation between new and returning users
- No product-level or category-level analysis
- Lack of marketing cost data limits profitability insights
- Assumes a linear user journey through the funnel

9. Conclusion

The analysis reveals that the main issue is not with the checkout process but with weak engagement at the top of the funnel and inconsistent traffic quality.

Improving early-stage user experience and focusing on high-performing acquisition channels can significantly enhance conversion rates and overall revenue.

10. Appendix A: SQL Queries

Funnel Analysis

WITH funnel_stages AS (

SELECT

COUNT(DISTINCT CASE WHEN event_type = 'page_view' THEN user_id END) AS stage_1_views,

COUNT(DISTINCT CASE WHEN event_type = 'add_to_cart' THEN user_id END) AS stage_2_cart,

COUNT(DISTINCT CASE WHEN event_type = 'checkout_start' THEN user_id END) AS stage_3_checkout,

COUNT(DISTINCT CASE WHEN event_type = 'payment_info' THEN user_id END) AS stage_4_payment,

COUNT(DISTINCT CASE WHEN event_type = 'purchase' THEN user_id END) AS stage_5_purchase

FROM user_events

)

SELECT * FROM funnel_stages;

Conversion Rates

SELECT

ROUND(stage_2_cart * 100 / stage_1_views) AS view_to_cart_rate,

ROUND(stage_3_checkout * 100 / stage_2_cart) AS cart_to_checkout_rate,

ROUND(stage_4_payment * 100 / stage_3_checkout) AS checkout_to_payment_rate,

ROUND(stage_5_purchase * 100 / stage_4_payment) AS payment_to_purchase_rate

FROM funnel_stages;

Revenue Metrics

SELECT

SUM(amount) AS total_revenue,

COUNT(*) AS total_orders,

ROUND(SUM(amount)/COUNT(*)) AS avg_order_value

FROM user_events

WHERE event_type = 'purchase';